

## **ASSIGNMENT 3 — Dynamic Dropdowns & Checkboxes**

### **1. Experiment Title**

#### **Handling Dynamic Dropdowns and Checkboxes Using Selenium**

### **2. Problem Statement**

Automate a web form containing multiple checkboxes and a search-as-you-type autocomplete dropdown. Select the required checkboxes, verify their selected state, enter text into the autocomplete field, and select the matching suggestion from the dynamically generated results.

### **3. Objective**

- To interact with checkboxes using Selenium.
- To verify checkbox state using `.is_selected()`.
- To interact with autocomplete fields.
- To iterate through dynamically generated suggestions.
- To select a matching dropdown option.

### **4. Tools, Software and Concepts Used**

#### **Tools & Software:**

- Python
- Selenium WebDriver
- Google Chrome
- VS Code / PyCharm

#### **Concepts:**

- Checkboxes
- `.is_selected()`
- Autocomplete Dropdown
- Dynamic Suggestions
- `find_elements()`

- Loops
- XPath
- Selenium WebDriver

## 5. Source Code / Implementation Steps

```
from selenium import webdriver

from selenium.webdriver.common.by import By

import time

def test_checkboxes_and_autocomplete():

    driver = webdriver.Chrome()

    driver.maximize_window()

    driver.get("https://www.selenium.dev/selenium/web/web-form.html")

    checkbox1 = driver.find_element(By.ID, "my-check-1")

    checkbox2 = driver.find_element(By.ID, "my-check-2")

    if not checkbox1.is_selected():

        checkbox1.click()

    if not checkbox2.is_selected():

        checkbox2.click()

    assert checkbox1.is_selected()

    assert checkbox2.is_selected()

    print("Checkbox 1 selected:", checkbox1.is_selected())

    print("Checkbox 2 selected:", checkbox2.is_selected())

    autocomplete = driver.find_element(By.NAME, "my-datalist")

    autocomplete.send_keys("New")

    suggestions = driver.find_elements(

        By.CSS_SELECTOR, "#my-options option"

    )
```

```
matching_option = None

for option in suggestions:

    if option.get_attribute("value") == "New York":

        matching_option = option.get_attribute("value")

        break

assert matching_option == "New York"

autocomplete.clear()

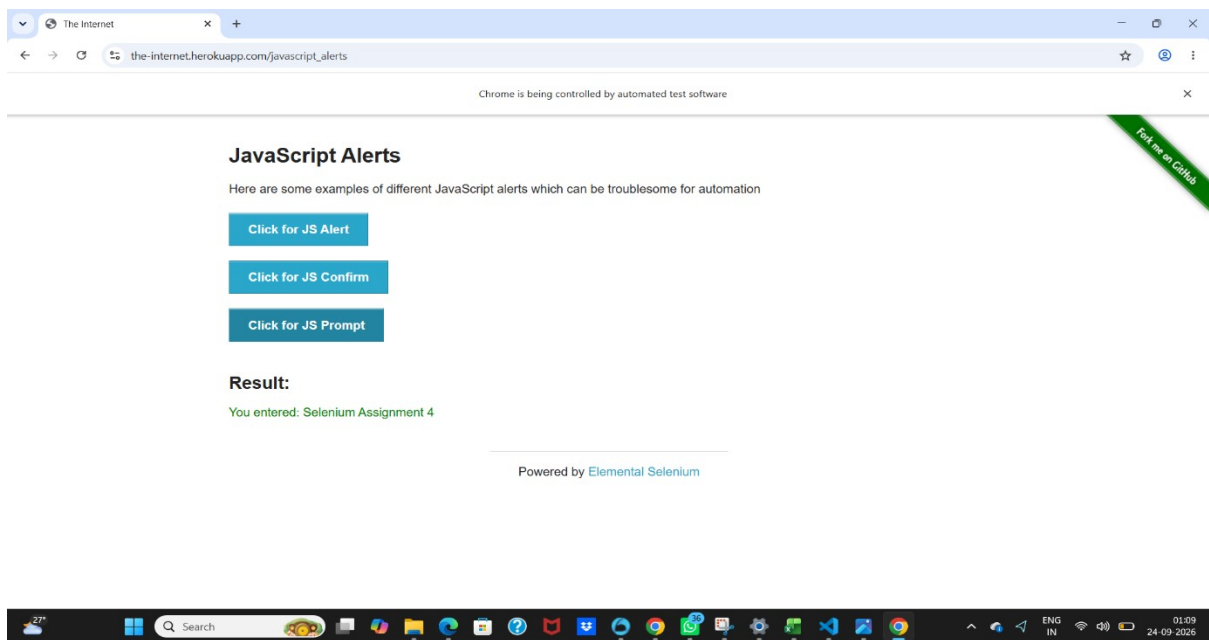
autocomplete.send_keys(matching_option)

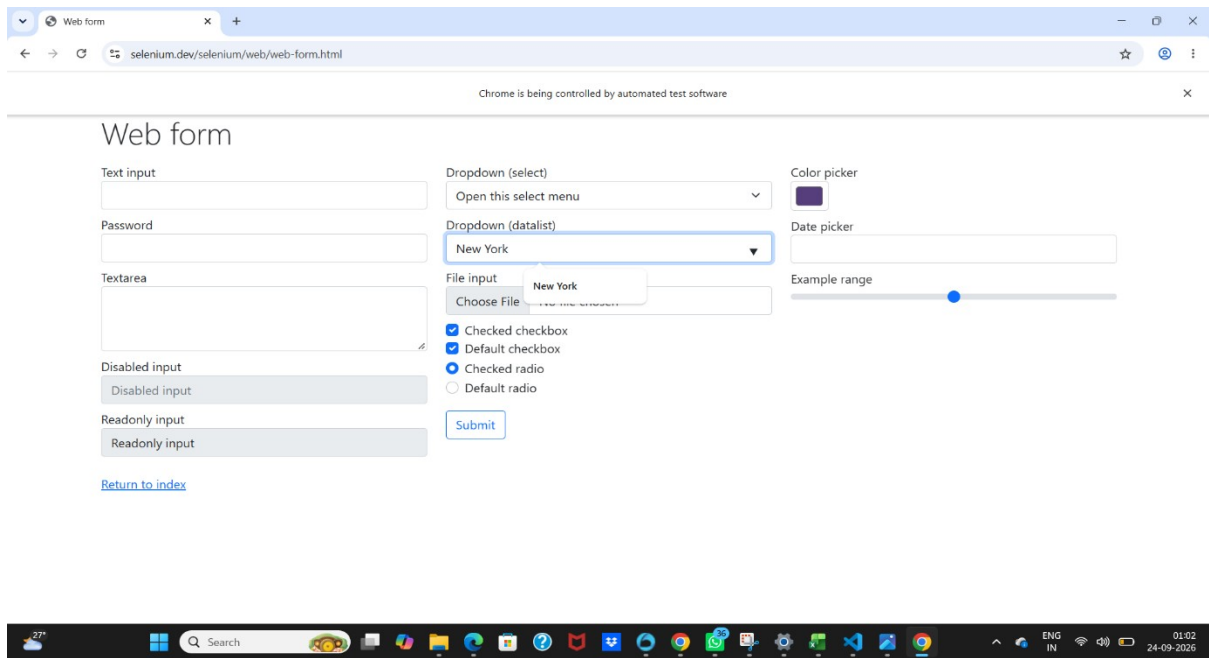
print("Selected autocomplete option:", matching_option)

time.sleep(10)

driver.quit()
```

## 6. Output





## 7. Result

The required checkboxes were successfully selected and their states were verified using `.is_selected()`. The autocomplete field was populated and a matching suggestion was selected from the dynamically generated results.

## 8. Observation

- `.is_selected()` verifies whether a checkbox is currently selected.
- Dynamic suggestions can be retrieved using `find_elements()`.
- Iterating through the suggestions allows the required option to be identified.
- Explicit waits can be used when suggestions are dynamically loaded.

## 9. Conclusion

This experiment demonstrated handling of **checkboxes and dynamic autocomplete dropdowns** using Selenium WebDriver, including state verification and selection of dynamically generated options.